GGG-S-131e
October 12, 1965

SUPERSEDING
Int. Fed. Spec. GGG-S-00131d (GSA-FSS)
December 16, 1962 and
Fed. Spec. GGG-S-131c
January 19, 1954

FEDERAL SPECIFICATION

SCRIBERS, MACHINISTS'

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** Machinists' scribers covered by this specification are designed for marking lines on steel and other materials in layout work.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 **Classification.**

1.2.1 *Types and classes.* Scribers shall be of the following types and classes as specified (see 6.1):

Type I—Single point, pocket.
  Class 1—Nonrolling head.
Type II—Double point, straight and bent.
  Class 1—Adjustable sleeve.
  Class 2—Removable.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

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3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of scribers otherwise conforming to this specification.

3.2 Materials. The body shall be of steel or aluminum alloy, and the scribing points shall be of hardened tool steel suitable for the purpose intended, or tungsten carbide when specified (see 6.1).

3.3 Components. Scribers shall consist of a body and a scribing point, except for type II, class 1, which utilizes a sleeve rather than a body.

3.3.1 *Body.* Body shall be designed to afford a comfortable handgrip. Types I and II bodies and sleeves shall be suitably knurled.

3.3.2 *Scribing points(s).* The scribing point(s) shall be of round steel or tungsten carbide as specified (see 6.1), having a gradual taper point(s).

3.4 Hardness of points.

3.4.1 *Steel.* The hardness of the point(s), and for a distance of at least 1/2 inch from the point(s), shall be not less than 55 nor more than 62 on the Rockwell C scale. The scribing point(s) shall show no breakage or deformation when tested in accordance with 4.4.1 and 4.4.1.1.

3.4.2 *Tungsten carbide.* Tungsten carbide point(s) shall have a hardness of 91 to 98 on the Rockwell A scale. Point(s) shall show no deformation or breakage when tested in accordance with 4.4.2.

3.5 Finish and coating. The entire surface of each scriber, except the knurled portions of the bodies or sleeves, shall be smooth and polished. Sharp corners shall be either beveled or rounded. Ferrous metal handles shall be plated or otherwise treated to prevent corrosion.

3.6 Marking. Scribers shall be stamped or otherwise permanently marked in a plain and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.7 Type I, single point, pocket. Type I scriber body shall hold the scribing point securely in position by means of a knurled chuck. The end of the scribing point, opposite the tapered point, may have a narrow groove near the end to simplify extraction from the body when the scribing point is sheathed. The scribing point shall be reversible, telescoping into the body as a protective measure.

TABLE I. *Type I, class 1, single point, pocket, nonrolling head*

Size	Diameter of handle	Length of handle		Total length with exposed scribing point		Diameter of scribing point body
		Minimum	Maximum	Minimum	Maximum	
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*
1	1/4	3-3/8	3-5/8	4-3/4	5-1/2	1/8
2	3/8	3-5/8	4	5-3/4	6-1/4	3/16

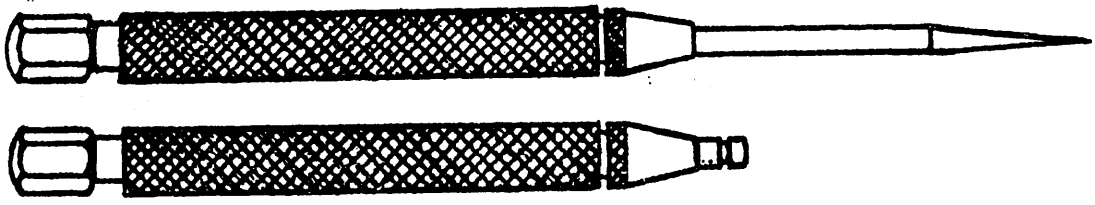


FIGURE 1. Type I, class 1, single point, pocket, nonrolling head

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FIGURE 2. Type II, class 1, double point, straight and bent, adjustable sleeve

3.7.1 *Class 1, nonrolling head.* Class 1 scriber shall terminate with a head not more than 9/16 inch in length either hexagonal or octagonal in cross section. The distance across the flats shall be not less than the diameter of the knurled portion of the body to prevent the scriber from rolling. Class 1 scribers shall conform to table I and be similar to figure 1. When specified (see 6.1), the scriber shall be furnished with a tungsten carbide point.

3.8 **Type II, double point, straight and bent.** Type II scribers shall have one straight and one bent tapered point. The straight tapered point shall coincide with the axis of the round body of the scribing point, and the bent tapered point shall be at 90° angle from the axis of the round body of the scribing point. The length of the bent tapered point shall be not less than 5/16 nor more than 1/2 inch from the round body of the bar.

3.8.1 *Class 1, adjustable sleeve.* Class 1 scriber shall have a scribing point of one-piece construction and a knurled adjustable sleeve which shall be a neat slip fit and adjustable along the body of the scribing point. The adjustable sleeve shall be provided with a chuck in which to secure the sleeve at any desired location along the body of the scribing point. Class 1 scribers shall conform to table II and be similar to figure 2.

TABLE II. *Type II, class 1, double point, straight and bent, adjustable sleeve*

Diameter of sleeve	Length of sleeve	Diameter of scribing point body (± 0.005 inch)	Length of scribing point
Inch	*Inches*	*Inch*	*Inches*
5/16	2-1/2 to 3	5/32	8 to 9

3.8.2 *Class 2, removable.* Class 2 scriber shall have two removable scribing points. The scribing points shall be suitably threaded to engage the threaded holes in the body. The body, when assembled to the scribing points, shall be midway in the scriber. When an extra long scribing point with a bent tapered point is required for use with class 2 scriber, it shall be specified (see 6.1). Class 2 scribers shall conform to table III and be similar to figure 3.

3.9 **Workmanship.** Workmanship shall be of the highest grade throughout, and equal in every respect to good commercial practice. The scribers shall be free from rust, fins, and other defects which may impair their serviceability and durability.

Diameter of handle	Length of handle	Diameter of scribing point body	Total length with short bent point (assembled)	Total length with both points (assembled)
Inch	*Inches*	*Inch*	*Inches*	*Inches*
1/4 to 5/16	2-3/4 to 3-1/4	1/8 to 3/16	8 to 9	12 to 13

TABLE III. *Type II, class 2, double point, straight and bent, removable.*



FIGURE 3. Type II, class 2, double point, straight and bent, removable

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table IV.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, workmanship, and marking.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.1.1 and 4.4.2.

4.4.1 *Performance test for steel points.* A metal plate (approximately 6 by 12 inches) with a hardness value of not less than 30 nor more than 36 on the Rockwell C scale, with top and bottom surfaces smoothly machined, shall be placed in a platform scale. The scriber under test shall be held by hand with the point on the plate while a pressure is exerted on the scriber until the scale indicates a reading of 4 pounds. The scriber shall be drawn under the 4-pound pressure, and on a natural scribing angle of inclination, for a distance of approximately 12 inches. One 12-inch test under the above conditions shall suffice. Under this test there shall be no perceptible increase in the width of lines scribed, nor shall there be any evidence of breakage or deformation of the points when examined under a microscope having a magnification not less than 50 power.

4.4.1.1 *Performance retest.* If a scriber has passed the test specified in 4.4.1, 1/4 inch of the straight working point shall be cut off. A new point shall be ground using ample coolant so as not to change the temper of the point. The test specified in 4.4.1 shall be repeated.

4.4.2 *Performance test for tungsten carbide points.* Each sample scriber with tungsten carbide point shall be tested by scribing a line on hard glass, such as window glass, for a distance of 24 inches. No breakage or deformation shall be evident when examined under a microscope having a magnification of not less than 50 power.

TABLE IV. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	4.0	Defects per hundred units	4.3.1
Dimensional examination ...	1 each	S-4	1.5	Defects per hundred units	4.3.2
Testing	1 each	S-2	1.0	Percent defective	4.4
Preparation for delivery	1 container	S-2	4.0	Defects per hundred units	4.5

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4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship .	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. The scribers shall be preserved and packaged level A or C, and packed levels A, B, or C of MIL-H-15424. The level of protection shall be as specified in the contract or order (see 6.1).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of scriber required (see 1.2).
(*c*) If scriber is to be furnished with tungsten carbide point (see 3.2, 3.3.2, and 3.7.1).
(*d*) Extra long scribing point with bent tapered point, when required (see 3.8.2).
(*e*) Levels of preservation, packaging, and packing required (see 5.1).
(*f*) Special marking, if required (see 5.3).

6.2 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A requirements in 5.1 shall apply.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

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